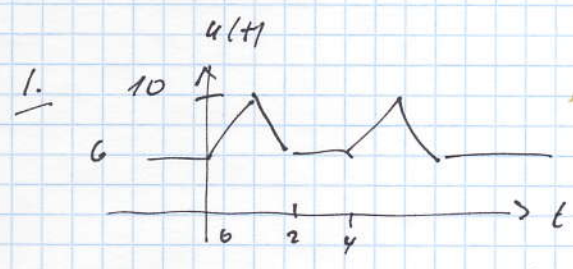
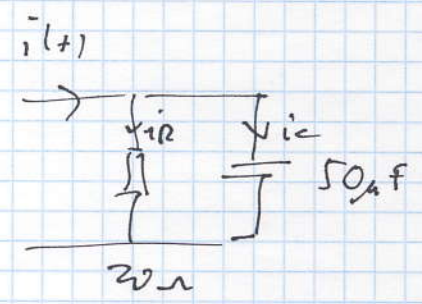


KSP, act 1, Februar 2015, 16a, 16b



$T=4$

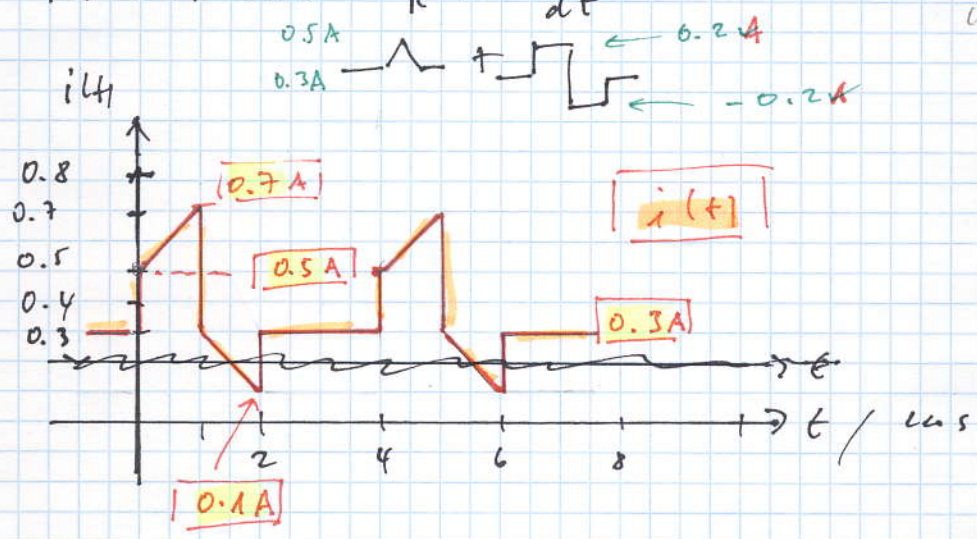


a) $\overline{u(t)} = \frac{2 \cdot 4 \cdot \frac{1}{2} + 6 \cdot 2 \cdot 2}{4} = 7V$

(2)

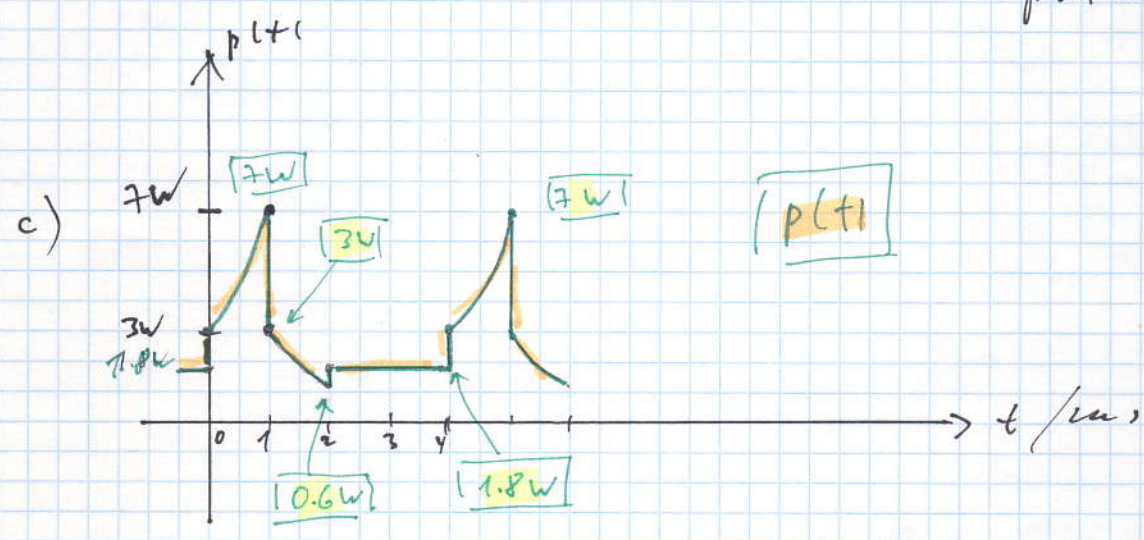
b) $i(t) = i_R + i_C = \frac{u}{R} + \frac{du}{dt} \cdot C$

F4 EGA
UR + BW



(4)

$p(t) = u(t) \cdot i(t)$



(4)

Fläche geschätzt: $3W \cdot 1ms + \frac{4W \cdot 1ms}{2} + 1.8W \cdot 1ms + 1.8W \cdot 2ms = 10.4 \mu Ws$

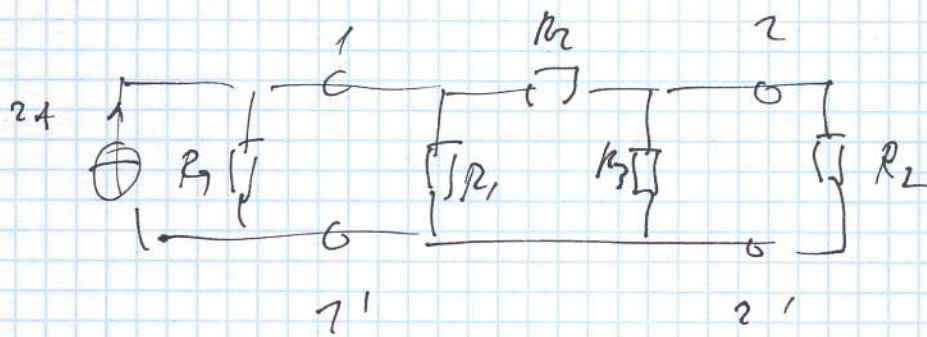
$\overline{p(t)} = \overline{P} = \frac{10.4 \mu Ws}{4ms} = 2.60 W = \frac{U_{eff}^2}{R}$

Erkelt: 2.533 W
($U_{eff} = 7.12V$)

$\hookrightarrow U_{eff} = 7.21V$

(2)

2.



$$\begin{aligned} R_1 &= 50 \\ R_2 &= 100 \\ R_3 &= 40 \\ R_L &= 220 \end{aligned}$$

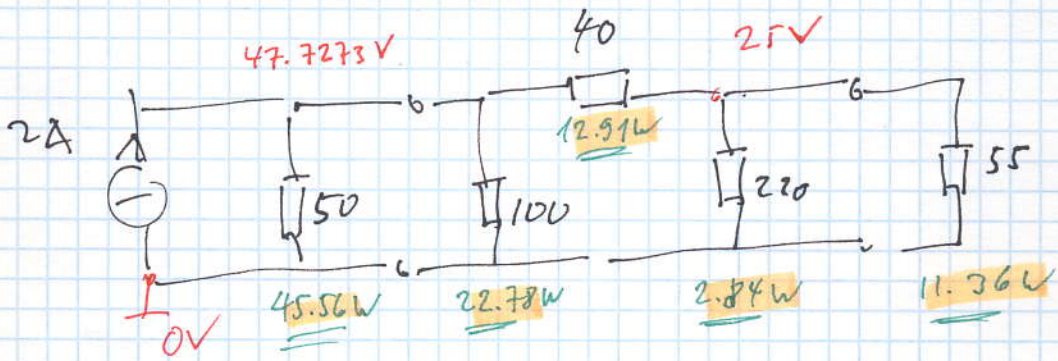
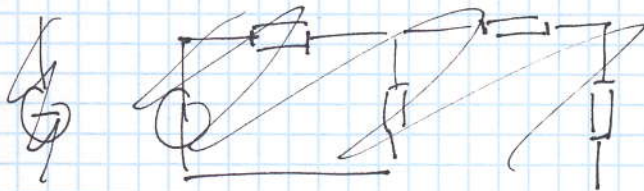
(2)

a) 1-1': $R_T = R_1 \parallel (R_2 + R_3 \parallel R_L) \Rightarrow \underline{\underline{R_T = 82.50 \Omega}}$

b) 2-2': $R_L = R_3 \parallel (R_2 + R_1 \parallel R_T) \Rightarrow \underline{\underline{R_L = 55.0 \Omega}}$

(2)

c)



(2)

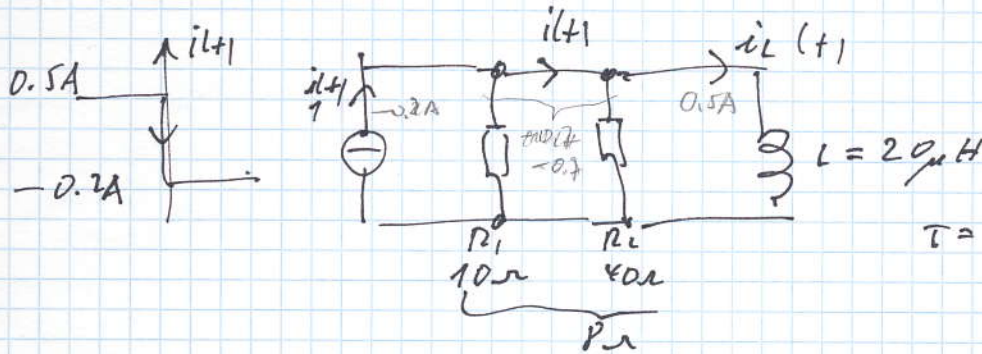
pro falsches Ergebnis,
- 0.5

~ L

3.

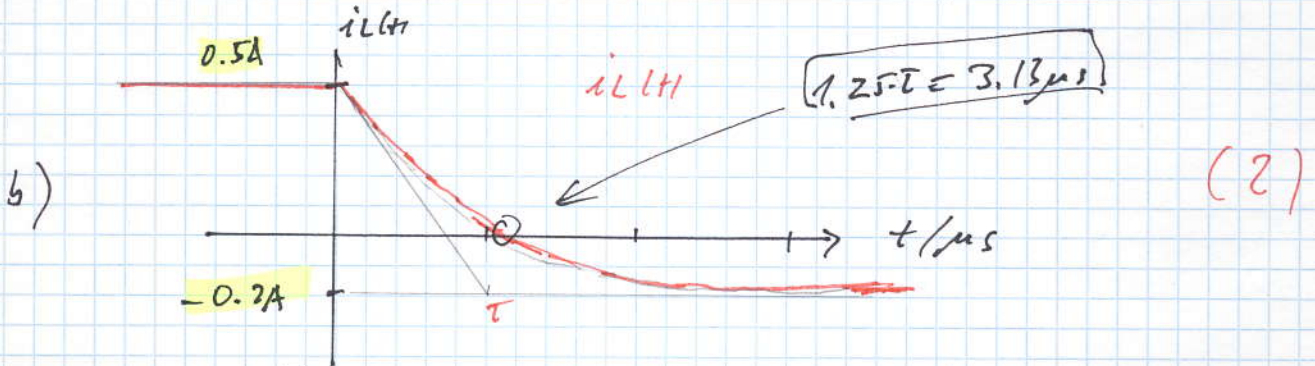
$U \rightarrow I$ -Quelle umwandeln

3



$$\tau = \frac{L}{R} = 2.5 \mu s$$

mit $i_L(t)$ beginn: $i(t \rightarrow 0) = 0.5A$
 $i(t \rightarrow \infty) = -0.2A$ } springt nicht



$$u_L(t) = L \cdot \frac{di}{dt}$$

$$u_L(t \rightarrow 0) = 0$$

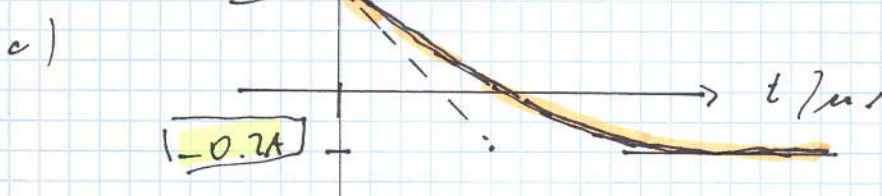
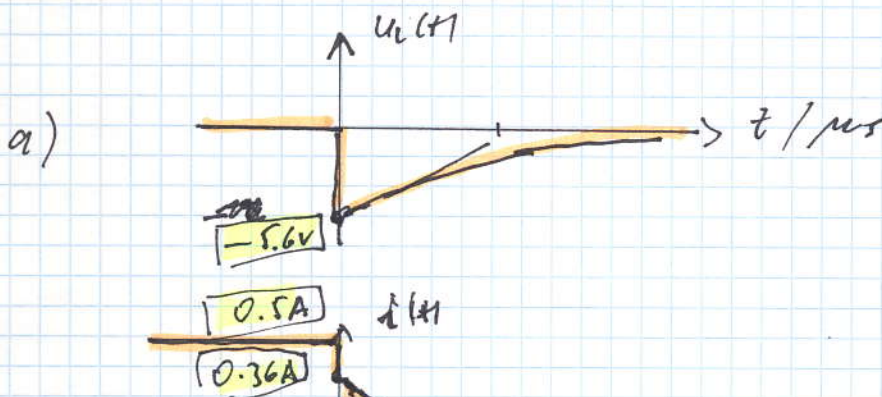
$$u_L(t \rightarrow \infty) = 0$$

$$t \rightarrow 0 \quad i_L = 0.5A$$

$$i_R = -0.2A$$

$$\rightarrow i_R = 0.7A \rightarrow$$

$$P_R = 5.6V$$



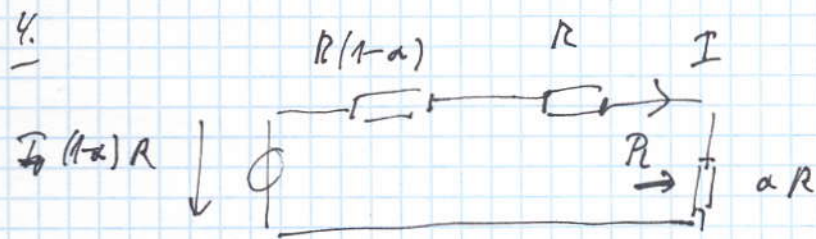
$$i_L(t) = 0.5 + (-0.7)(1 - e^{-t/\tau}) \stackrel{!}{=} 0$$

$$\Rightarrow t_0 = 1.25\tau = 3.13\mu s$$

d)

$$P_2 = \frac{u_{R2}^2}{R_2} = u_0^2 \cdot e^{-2t/\tau} \cdot \frac{1}{R_2}$$

$$W = \frac{u_0^2}{R_2} \cdot \int_0^\infty e^{-\frac{2t}{\tau}} dt = 0.98\mu J$$



a)
$$I = \frac{I_0(1-\alpha)R}{R - \cancel{\alpha R} + R + \cancel{\alpha R}} = \frac{I_0 \cdot \cancel{R}}{2} \cdot (1-\alpha)$$

(2)

b)
$$P_2 = I^2 \cdot \alpha R = \frac{I_0^2 \cdot \cancel{R}}{4} \cdot \alpha \cdot R (1-\alpha)^2$$

$$P_2 = \frac{I_0^2 \cdot R}{4} \cdot \alpha \cdot (1-\alpha)^2$$

(2)

Nullo Stellen bei $\alpha_1 = 0$

$\alpha_2 = 1$
 $\alpha_3 = \frac{1}{3}$

$$\frac{dP_2}{d\alpha} = K \cdot [1 \cdot (1-\alpha)^2 + \alpha \cdot 2(1-\alpha) \cdot (-1)]$$

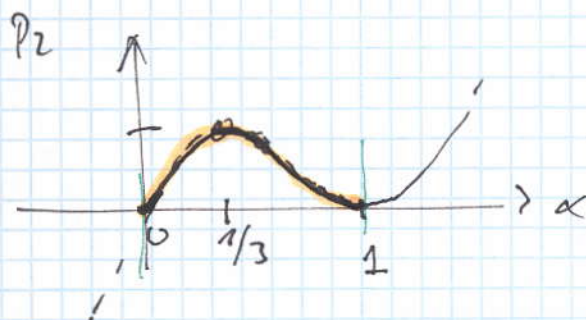
$$= K [\cancel{\alpha^2} - 2\alpha + 1 + \cancel{\alpha^2} - 2\alpha + 1]$$

$$= K [\alpha^2 - 2\alpha + 1 - 2\alpha + 2\alpha^2] = K [3\alpha^2 - 4\alpha + 1]$$

$$\frac{4 \pm \sqrt{16-12}}{6} = \left\{ \frac{1}{3}, 1 \right\}$$

Maxim bei $\alpha = \frac{1}{3} = 0.33$

(2)



(2)

5.

(5)

$$9) \quad \overline{i(t)} = \frac{\overline{I_2} \cdot 2 + \overline{I_1} \cdot 2 \cdot \frac{1}{2}}{4} = \frac{\overline{I_2}}{2} + \frac{\overline{I_1}}{4}$$

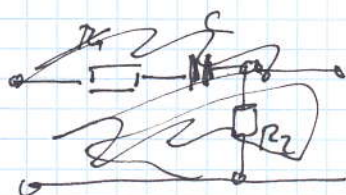
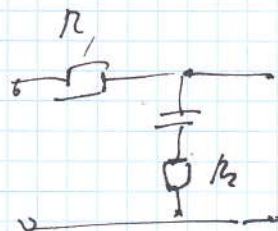
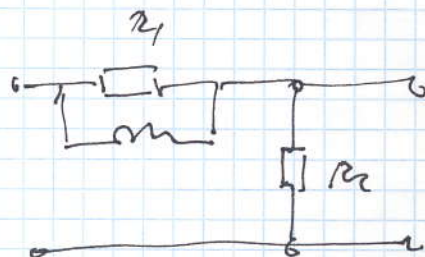
$$\overline{I_{eff}^2} = \frac{\overline{I_2^2} \cdot 2 + \frac{\overline{I_1^2}}{3} \cdot 2}{4} = \frac{\overline{I_2^2}}{2} + \frac{\overline{I_1^2}}{6}$$

6)

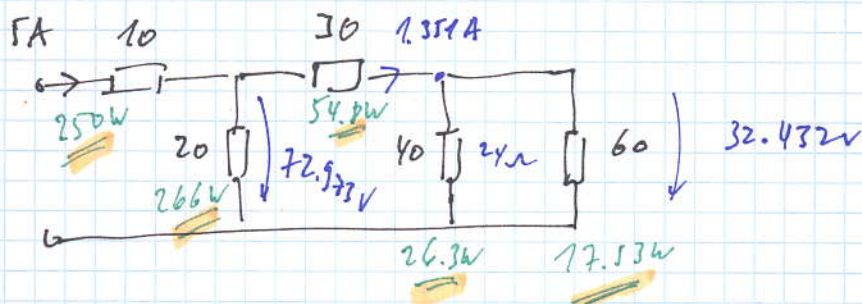
$$\underline{R_{ge}} = 40 + 10 = 90 \, \Omega$$

$$\underline{I_{ge}} = 0.1 \text{ A} + \frac{15 - 10}{90} = 0.152 \text{ A}$$

$$\underline{U_{ge}} = 14 \text{ V}$$

c) Für DC $\rightarrow V_{cont} = 1$ ~~Diagram~~

d)



e)

$$\underline{R_{ge}} = \frac{0.4}{0.1} = \frac{0.2 \text{ V}}{6.7 \text{ A}} = 0.030 \, \Omega$$

$$\underline{U_{ge}} = \dots = 13.44 \text{ V}$$